Name: Soundariya M

Reg No: 192524342

Course Code: DSA0401

Course Name: FUNDAMENTAL OF DATASCIENCE

Question 5: Average Fuel Efficiency

AIM:

To calculate average fuel efficiency and percentage improvement between two car models using NumPy.

ALGORITHM:

1. Import NumPy.

2. Store fuel efficiency values in an array.

3. Calculate average using np.mean().

4. Calculate percentage improvement between two models.

5. Display the results.

CODE:

```
import numpy as np

fuel_efficiency = np.array([25, 30, 35, 40])
```

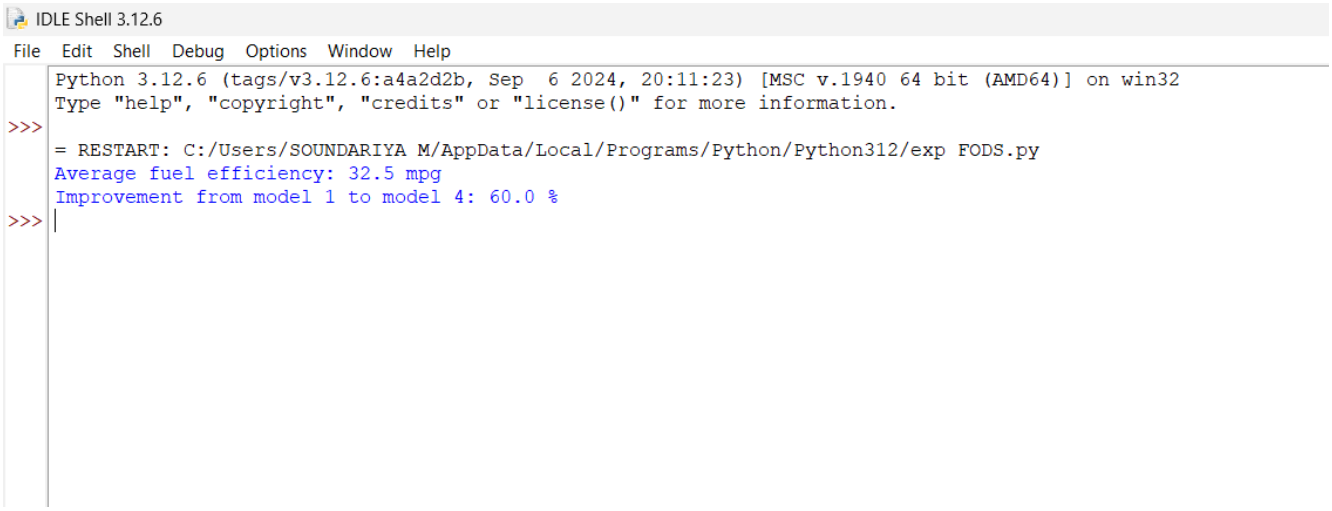
```
average = np.mean(fuel_efficiency)
old_model = fuel_efficiency[0]
new_model = fuel_efficiency[3]
improvement = ((new_model - old_model) / old_model) * 100

print("Average fuel efficiency:", average, "mpg")
print("Percentage improvement:", improvement, "%")
```

INPUT:

fuel_efficiency = [25, 30, 35, 40]

OUTPUT:



RESULT:

The average fuel efficiency is 32.5 mpg, and improvement from the first to fourth model is 60%.